

# Problem 3.11

BSA, 2024

```
In[1]:= ClearAll["Global`*"]  
Needs["VariationalMethods`"]  
$Assumptions={c>0,m>0,l>0,ω>0,L>0,r>0,g>0};
```

Lagrangian

Hamiltonian

First integrals

Equilibrium states

Linearization

Non-Dimensionalization

Stability domain

Crossing the border

## Solutions planes. Configuration space. Phase space

In[255]:=

```
X = {H[τ], Θ[τ], A[τ], B[τ]};
X0 = Thread[(X /. τ → 0) == {0.01, 0.01, 0, 0}];
ConvertToX = {A'[τ] → H'[τ], A'[τ] → H[τ], B'[τ] → Θ'[τ], B'[τ] → Θ[τ]};
ToTor[x_] := {(x[[4]] * Cos[x[[2]] + x[[3]]) * Sin[x[[1]]], (x[[4]] * Cos[x[[2]] + x[[3]]) * Cos[x[[1]]], x[[4]] * Sin[x[[2]]]}
```

In[259]:=

```
LinSys = Union[LaEqLinNonDim /. ConvertToX, {A'[τ] == H[τ], B'[τ] == Θ[τ]}];
AlmLinSys = Union[LaEqAlmLinNonDim /. ConvertToX, {A'[τ] == H[τ], B'[τ] == Θ[τ]}];
NonLinSys = Union[LaEqNonDim /. ConvertToX, {A'[τ] == H[τ], B'[τ] == Θ[τ]}];
```

## Linear system

## Full analysis

In[264]:=

```

Solution[g_] := (
  t = If[g > -0.245, 100, 17];
  ConvertToGT = { $\gamma \rightarrow (1 - g) / 2$ ,  $\sigma \rightarrow (g + 1) / 2$ ,  $\epsilon \rightarrow 4$ };

  Sollin = NDSolveValue[Union[LinSys /. ConvertToGT, X0], X, { $\tau$ , 0, t}];
  SolAlmLin = NDSolveValue[Union[AlmLinSys /. ConvertToGT, X0], X, { $\tau$ , 0, t}];
  SolNonLin = NDSolveValue[Union[NonLinSys /. ConvertToGT, X0], X, { $\tau$ , 0, t}];

  pp1 = ParametricPlot3D[{ToTor[{Sollin[[3]], Sollin[[4]], 2 + Sollin[[1]], 1 + Sollin[[2]]}],
    ToTor[{SolAlmLin[[3]], SolAlmLin[[4]], 2 + SolAlmLin[[1]], 1 + SolAlmLin[[2]]}],
    ToTor[{SolNonLin[[3]], SolNonLin[[4]], 2 + SolNonLin[[1]], 1 + SolNonLin[[2]]}]], { $\tau$ , 0, t},
    PlotStyle → {Blue, Red, Darker[Green]}, PlotLegends → Placed[{"Lin", "AlmLin", "NonLin"}, Bottom], PlotRange → Full];
  pp2 = ParametricPlot3D[ToTor[{u, v, 2, 1}], {u, 0, 2*Pi}, {v, 0, 2*Pi}, PlotStyle → {Gray, Opacity[0.2]}];

  p1 = Plot[{Sollin[[3]], SolAlmLin[[3]], SolNonLin[[3]]}, { $\tau$ , 0, t}, AxesLabel → { $\tau$ , A[ $\tau$ ]}, GridLines → Automatic, PlotRange → All,
    PlotStyle → {Blue, Red, Darker[Green]}, PlotLegends → Placed[{"Lin", "AlmLin", "NonLin"}, Bottom], ImageSize → Full, AspectRatio → 9/10];
  p2 = Plot[{Sollin[[4]], SolAlmLin[[4]], SolNonLin[[4]]}, { $\tau$ , 0, t}, AxesLabel → { $\tau$ , B[ $\tau$ ]}, GridLines → Automatic, PlotRange → All,
    PlotStyle → {Blue, Red, Darker[Green]}, PlotLegends → Placed[{"Lin", "AlmLin", "NonLin"}, Bottom], ImageSize → Full, AspectRatio → 9/10];
  p3 = ComplexListPlot[Table[( $\lambda$  /. EigVal[[i]]) /. ConvertToGT, {i, Length[EigVal]}], PlotRange → {{-2, 2}, {-2, 2}},
    PlotStyle → {PointSize[Large], Red}, AxesLabel → {"Re", "Im"}];
  p4 = Show[RegionPlot[StabDomainCond, { $\gamma$ , 0, 1}, { $\sigma$ , 0, 1}], Graphics[{Red, PointSize[Large], Point[{(1 - g) / 2, (1 + g) / 2}]}];
  p5 = Show[pp1, pp2, SphericalRegion → True, ImageSize → Full];
  p6 = ParametricPlot[{Sollin[[3]], Sollin[[4]]}, {SolAlmLin[[3]], SolAlmLin[[4]]}, {SolNonLin[[3]], SolNonLin[[4]]}, { $\tau$ , 0, t},
    AxesLabel → {A, B}, GridLines → Automatic, PlotRange → All, PlotStyle → {Blue, Red, Darker[Green]},
    PlotLegends → Placed[{"Lin", "AlmLin", "NonLin"}, Bottom], ImageSize → Full, AspectRatio → 9/10];

  GraphicsGrid[{{p5, p3, p1}, {p6, p4, p2}}, ImageSize → Large]
);
Animate[Solution[G], {G, -0.33, 0.1, Appearance → "Labeled"}, AnimationRunning → False]

```

Out[265]=

